UTP Universiti Teknologi PETRONAS — emerging futures

GRAND FINALE · HEALTHTECH TRACK

CODEFEST'26 - SYNTECH -

Kawan.

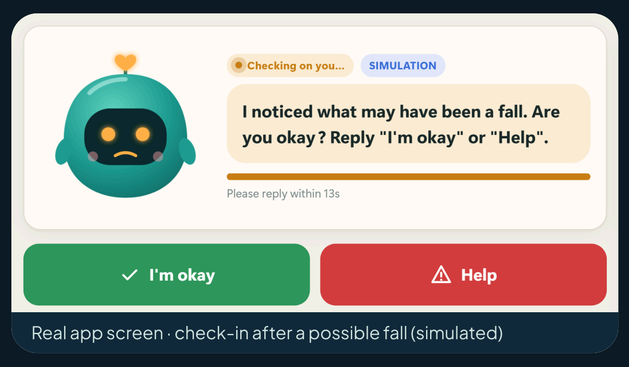
kawan (Malay) · *friend*

An AI guardian for people who live alone: it notices, checks in, and alerts family only when needed.

Not a doctor. Not a surveillance camera. A guardian.

Team Nexora UTP CODEFEST '26

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Real app screen · check-in after a possible fall (simulated)

01 The problem

IMPACT

15.6% of Malaysian adults have **diabetes**, 29.2% hypertension; ~4 in 10 don't know it.[1]

684k people die from **falls** each year; most fatal falls are in over-60s.[2]

80% of fallers over 90 **couldn't get up alone**; 30% lay on the floor ≥ 1 hour.[3]

~50% of long-term patients **don't take medicine** as prescribed.[4]

CONVENTIONAL ALARM · PENDANT OR PANIC BUTTON

Person falls → They must press the button → Hurt or confused: alarm mostly unused[3], no one knows

02 Our solution

IMPACT · USABILITY

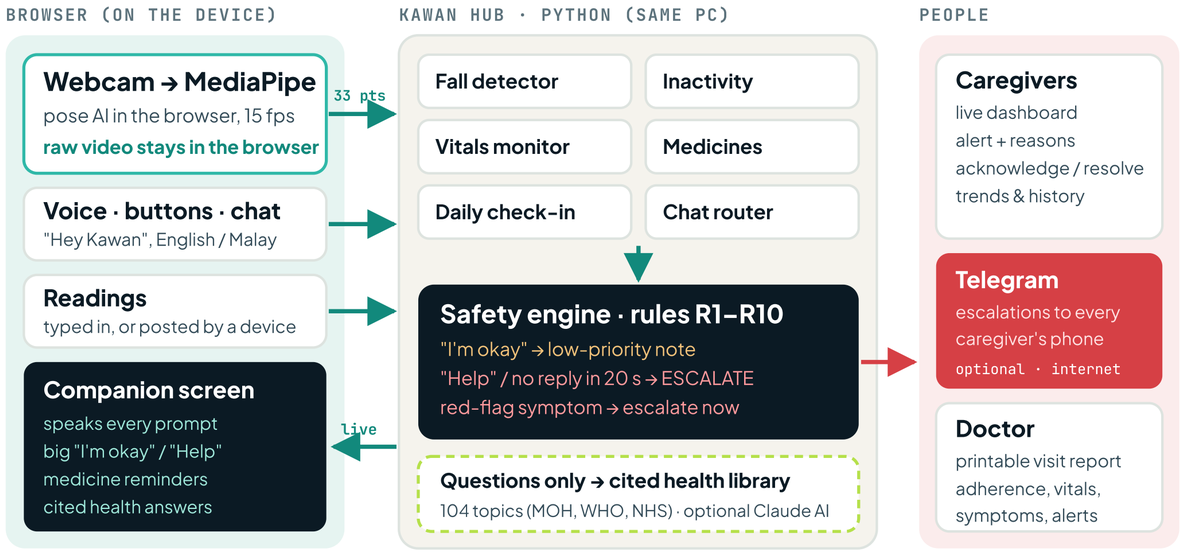
1. **Notice** Camera spots falls and unusual stillness; tracks medicines and readings.
2. **Check in** "**Are you okay?**" by voice, button or chat, in English or Malay.
3. **Escalate only if needed** "Help", a red-flag symptom or **no reply in 20 s** alerts caregivers.
4. **Explain why** Caregivers see **why**, and what else it could be.

KAWAN · PROACTIVE CHECK-IN · NO PANIC BUTTON, NO WEARABLE

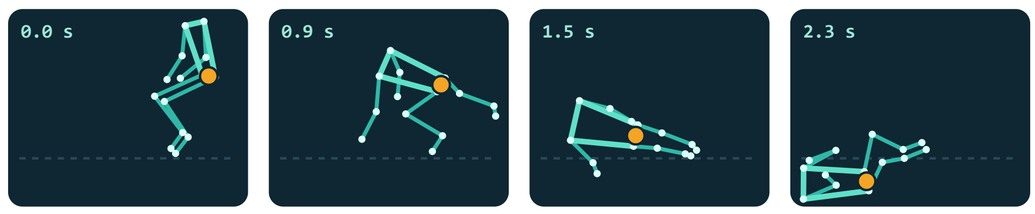
Webcam notices the fall → "Are you okay?" → "Help" or no reply in 20 s: caregivers alerted, with reasons

03 How it works

TECHNICAL EXECUTION



Fall detection on a recorded fall clip



upright	tipping over	hips dropping fast	on the floor
torso 11° · hip 0.60	torso 63° · hip 0.64	torso 72° · hip 0.84	torso 87° · hip 1.03

1. **Upright**, then
2. **hips drop fast:** ≥ 0.6 of their own torso length in ≤ 1.5 s
3. **body horizontal** (≥ 60°) or out of view
4. **still down** after 2 s → check in

```
Kawan's own output for this clip:
Hip dropped 14% of frame (0.6× torso) in 1.3 s · torso 90° · no recovery within 2 s
```

Skeleton extracted from UR Fall clip fall-04.[5]

04 The app

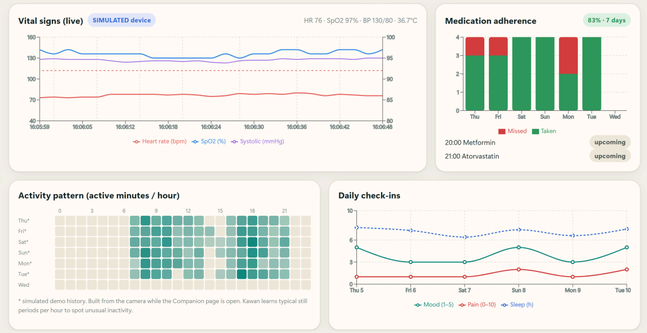
USABILITY · REAL SCREENS, DEMO DATA



Fall alert, with reasons
Why it fired, what else it could be.



On a phone
Bahasa Melayu



Caregiver dashboard
Vitals, adherence, activity, check-ins.

ALSO BUILT AND WORKING

- **Spoken reminders**, missed-dose notice
- **Vitals** with sustained-reading alerts
- **Daily check-in** triage, up to "call 999"
- **104 cited health topics**, EN & BM
- **Doctor report**, one click

05 Safe & private by design

- **Rules decide, AI only explains.** Check-in replies never go to the AI.
- **Fails safe.** "Okay, but my leg hurts" counts as help; unclear replies keep the timer running.
- **Explains itself:** every alert lists why it fired and what else it could be.

WHAT LEAVES THE BROWSER

Camera video **stays in the browser** → **33 landmarks** x, y, visibility · 15/s → Kawan hub **same PC by default**

No video or images are sent or stored; the hub keeps only events and summaries. **One opt-in exception:** "Check my medicine" sends a single photo to Claude when tapped (needs an AI key).

06 Offline & online

Works without internet
- Pose detection, fall & inactivity rules
- Check-ins by button and chat
- Spoken prompts (device voices)
- Reminders, vitals, daily check-in
- 104-topic health library
- Dashboard & report (on the hub PC)

Needs internet
- Voice input (browser speech service)
- Telegram alerts to phones
- Claude AI answers & pill check (optional, API key)
- Phones & remote caregivers (secure link)

07 Results

TECHNICAL EXECUTION · USABILITY

PRELIMINARY TEST · FALL DETECTOR ON RECORDED CLIPS[5]

27 / 30 **fall clips detected (90%)**. The earlier frame-size rule: 17/30.

12 / 40 **everyday clips triggered a check-in**; 10 of them show people lying down on purpose.

A check-in asks the person → "I'm okay" = low-priority note → Phones alerted only on "help" or no reply

Test conditions: UR Fall dataset, 70 lab clips of actors, one room, side-view camera; same pose model and settings as the app; each clip's last frame held 10 s (person stays down).

Not real-world accuracy. Preliminary lab testing; not clinically validated.

148 **automated tests**: 117 backend · 19 frontend · 12 browser end-to-end, run by GitHub Actions on every push.

EARLY INFORMAL FEEDBACK · 4 TESTERS (3 FRIENDS AND A PARENT)

> *"The reminders are the part I'd use every day."*
> *"I don't need to read anything. I just talk to it and it talks back."*
> *"It noticed when I fell and asked if I was okay."*
> *"I asked what ibuprofen is and finally understood it."*

WHAT WE HEARD	WHAT WE DID / WILL DO
Alerts on 2–3 caregivers' phones	✓ **Done:** Telegram to every caregiver
Voice sometimes misses words	Tap buttons as backup; better speech next
Knows less than chatbots	Deliberate: cited sources only

08 Feasibility & next steps

FEASIBILITY

Ready to run today

Hardware: a laptop or tablet with a webcam + Chrome/Edge.
Cost: free open-source stack; AI and Telegram optional.
Setup: one command; a secure link for phones.

Roadmap

NOW Working prototype; tried by 4 early users.

NEXT Caregiver login · hosted multi-household version · push alerts · device integration.

LATER Clinical validation with healthcare partners · more emergency types.

REFERENCES

1. IPH, MOH Malaysia. *NHMS 2023: NCD Fact Sheet.* 2024.
2. WHO. *Falls — Fact sheet.* 2021.
3. Fleming J, Brayne C. Inability to get up after falling (people over 90). *BMJ* 2008;337:a2227.
4. WHO. *Adherence to Long-Term Therapies.* 2003.
5. Kwolek B, Kepski M. *Comput Methods Programs Biomed* 2014;117:489–501 (UR Fall dataset, CC BY-NC-SA 4.0).
6. Google. *MediaPipe Pose Landmarker.*

Prototype — not a medical device. In an emergency call 999. One camera, one person in view. Health library from MOH Malaysia CPGs, WHO and NHS. Built with help from Claude Code (AI coding assistant), as the rules require us to disclose.

DEMO VIDEO SOURCE CODE